



Technology Transfer & Patent Rights

PPT PRESENTED BY

NAME : Srinivas Rao Tammireddy

ROLL NO : 257Y1D5805

YEAR : I Year I Semester

ACADEMIC YEAR : 2025 - 2026

Scope of Patent Rights

A Patent grants the owner exclusionary rights to prevent others from making, using, selling, or importing the invention. It represents a legal monopoly with distinct boundaries.

Legal Boundaries of Patents

- **Exclusionary Right:** Does not grant the owner the right to make the invention — it only grants the right to stop OTHERS from doing so. (Subject to regulatory and prior patent locks).
- **Territorial Scope:** Rights are valid only within the boundaries of the granting nation.
- **Term Limit:** Standard term is 20 years from the filing date, after which it enters the public domain.
- **Maintenance Fees:** Requires paying annual maintenance fees to keep the patent valid.

Licensing & Technology Transfer

- **Technology Transfer:** Sharing and commercializing innovations from research labs (universities) to industrial partners.
- **Licensing Models:** Legal contracts granting third parties rights to use the IP in exchange for royalties:
 - **Exclusive License:** Only one licensee has the rights.
 - **Non-Exclusive:** Multiple competitors can buy rights (e.g. video codecs).
- **Cross-Licensing:** Competitors trade patent portfolios to prevent mutual litigation.

Geographical Indications (GIs)

Geographical Indications protect products whose quality, characteristics, or reputation are due to their origin:

What is a GI?

- **Geographical Origin:** A label used on agricultural, natural, or manufactured goods that have a specific geographic origin.
- **Inherent Quality Link:** The product's characteristics must be due to the soil, climate, or traditional methods of that origin (e.g. Darjeeling Tea).
- **Collective Right:** Belongs to a community of local producers, not a single private company. Prevents misuse of the name by foreign competitors.
- **Duration:** Typically granted for 10 years, but can be renewed indefinitely.

GI Examples in India

India enacted the GIs of Goods Act in 1999 to protect local specialties:

- **Darjeeling Tea:** The first GI tag registered in India (2004). Protections prevent foreign growers from labeling inferior tea as Darjeeling.
- **Basmati Rice:** Aromatic long-grain rice grown in specific northern plains.
- **Pashmina Shawls:** Hand-crafted wool products from Kashmir.
- **Pochampally Ikat:** Traditional handloom sarees from Telangana.

New Developments in IPR

IP systems are adapting to protect digital, biological, and traditional innovations:

Biotechnology & Software IP

- **Biological Systems:** Patenting genetically modified organisms (like GM cotton seeds), gene editing protocols (CRISPR), and vaccine formulas. Raises ethical debates.
- **Software & AI IP:** AI algorithms are abstract math (non-patentable). However, systems where AI controls physical robots can be patented. Generative AI raises debates on copyright ownership of machine-generated code.

Traditional Knowledge (TK)

- **The Challenge:** Bio-pirating companies patenting ancient remedies (e.g. Turmeric, Neem) in foreign patent offices.
- **The India Case:** India fought and revoked US patents on Turmeric healing properties by presenting ancient Sanskrit texts as prior art.
- **TKDL Database:** Traditional Knowledge Digital Library translates ancient medical texts into 5 global languages, allowing patent offices to find prior art immediately and block invalid patents.

Applications of Technology Transfer

How technology transfer mechanics bridge the gap between academic labs and market sales:



University spin-offs

Universities set up Technology Transfer Offices (TTOs) to license patents created by professors and students to startup companies, commercializing research.



Vaccine Licensing Deals

Research institutes (like Oxford University) patent vaccine formulas and license manufacturing rights to global firms (like AstraZeneca) for production.



Agricultural Licensing

Agricultural universities license disease-resistant seed strains to farming cooperatives, increasing crop yields through research.



Open-source patent pledges

Tech companies pledge patent portfolios to open-source consortia, allowing startups to build services without litigation risks.



THANK YOU

Department of Computer Science and Engineering
Marri Laxman Reddy Institute of Technology and Management